INCH-POUND
MIL-DTL-24643/39E
22 August 2002
SUPERSEDING
MIL-C-24643/39D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1S75MU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, tin coated, ASTM B 286, AWG Size No. 22,7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.070 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth -	Braided shield of AWG No. 34 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.
Sixth -	The eight singles cabled together with a lay not greater than 8 inches around a center filler. Cabling sequence shall be consecutive, starting with No. 1, from the center outward.

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Construction (non-watertight) (continued)

Seventh - Binder tape, applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.050 inch.

TABLE I. Details.

Military part no. 1/ M24643/39	Size	Number of Conductors	Diameter over completed conductor (inch) (nominal)	Overall diameter (inches) TYPE LS1S75MU	
				min	max
-01UN	8	8	0.228	0.950	1.030

1/ When double shield is required, see MIL-DTL-24643 for configuration of part number.

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, maximum 17.37
Voltage withstand - volts, root mean square, minimum
Conductor to shield 7000
Shield to shield (component shields only) 500
Insulator resistance – Megohms-1000 feet minimum
Conductor to shield 500
Component shield to shield 100
Conductor continuity ..... No failure

Group A:
Visual and dimensional ..... No failure
Characteristic impedance - at 1MHz, ohms ..... 75 ± 3

Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When test at 200°C) 50
Attenuation - at 400 MHz, decibel (dB)/100 ft maximum 9.0
Cold bending, cable - at 0 ± 2°C, (see table I for mandrel diameter) ..... No damage
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

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Group C:
Physicals - (aged)
- Insulation
  - Air Oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air Oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5

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Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-037)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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